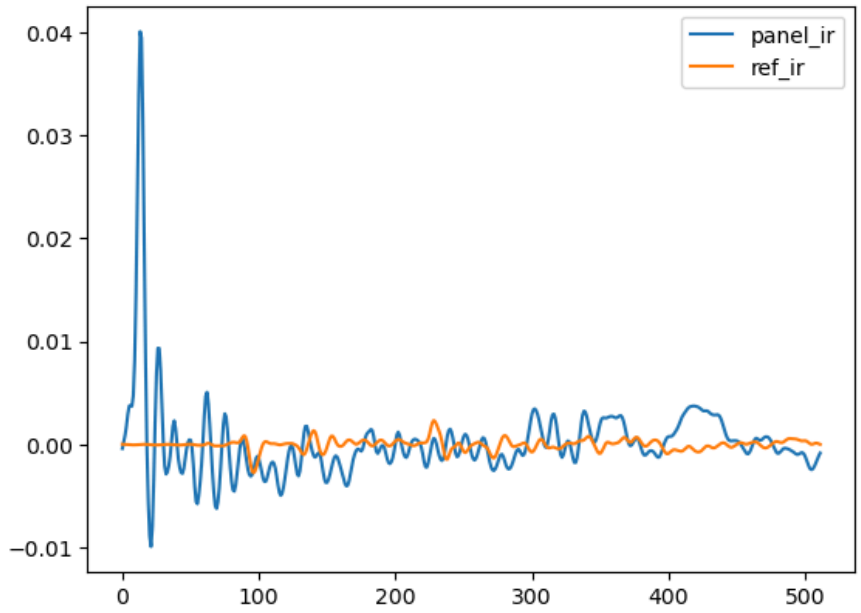
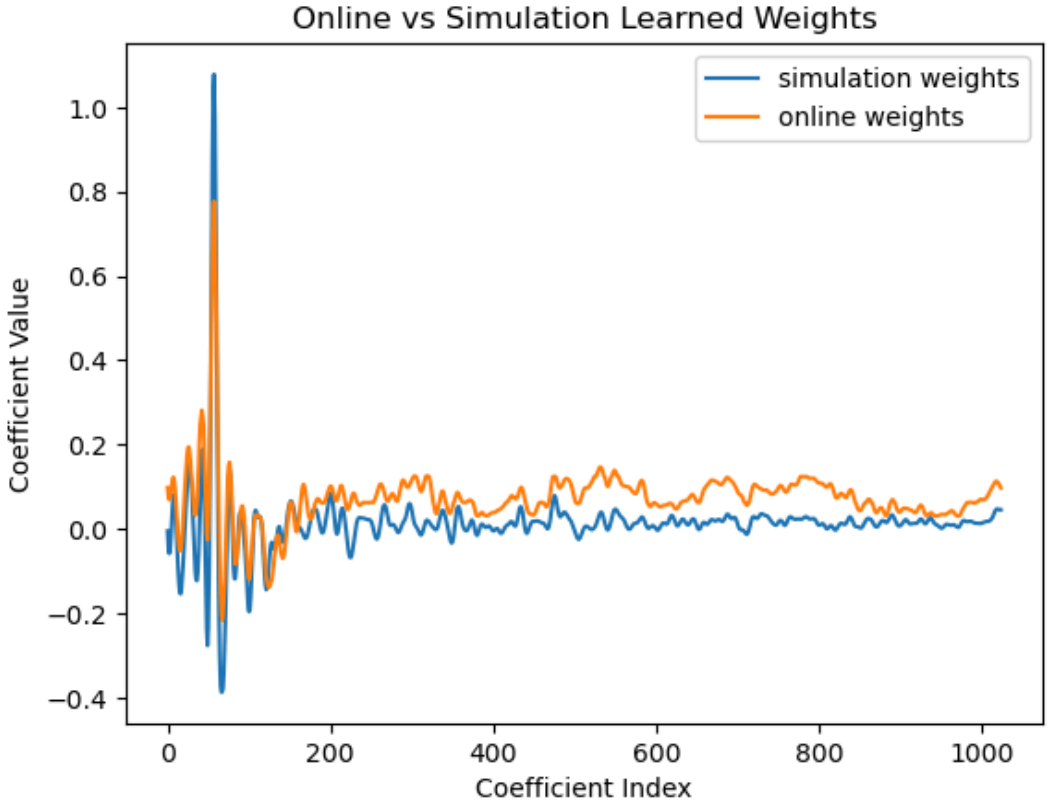
Added ref feedback to the simulation

- Barely affected anything
- The ref IR starts later than the error IR, but that extra tail doesn't really contain anything

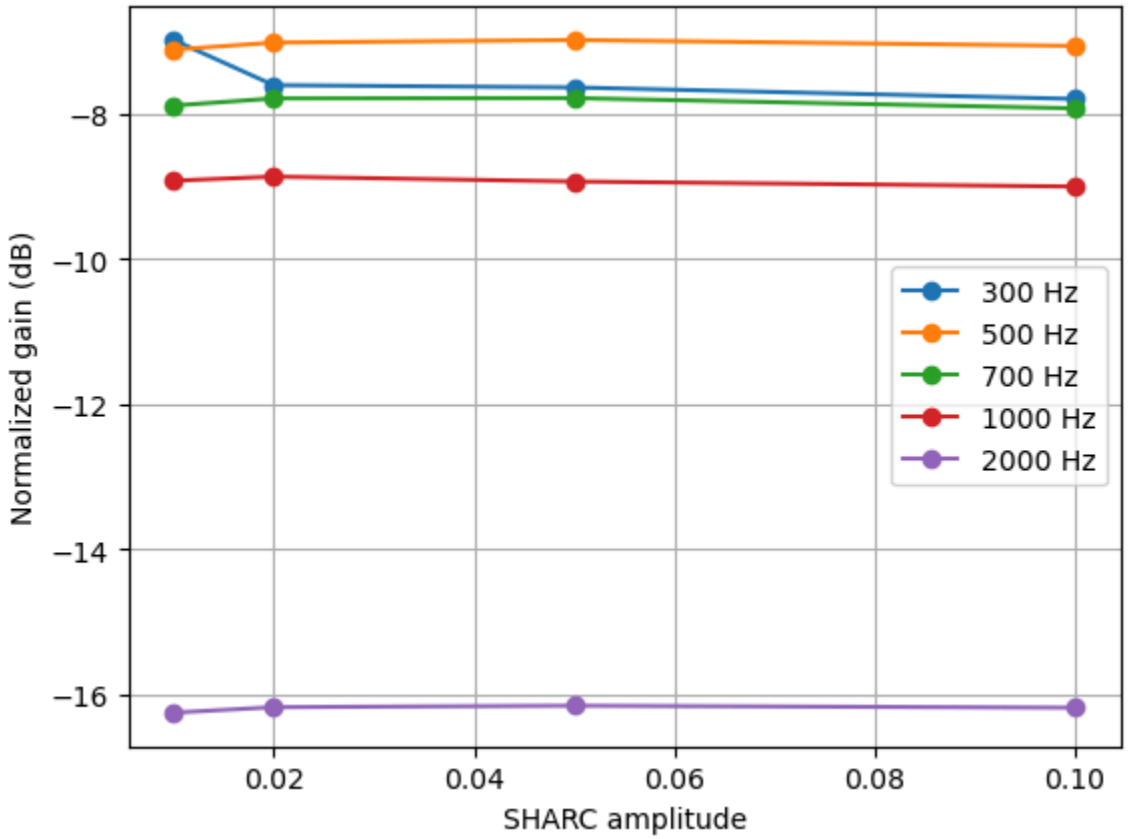


Online weights vs simulation weights



Grid searched pure tones for linearity

- Record a tone at a specific frequency and amplitude
- Compute the complex Fourier component of the recording at that frequency and take the magnitude
- Normalized gain is the dB conversion of that magnitude divided by the original amplitude

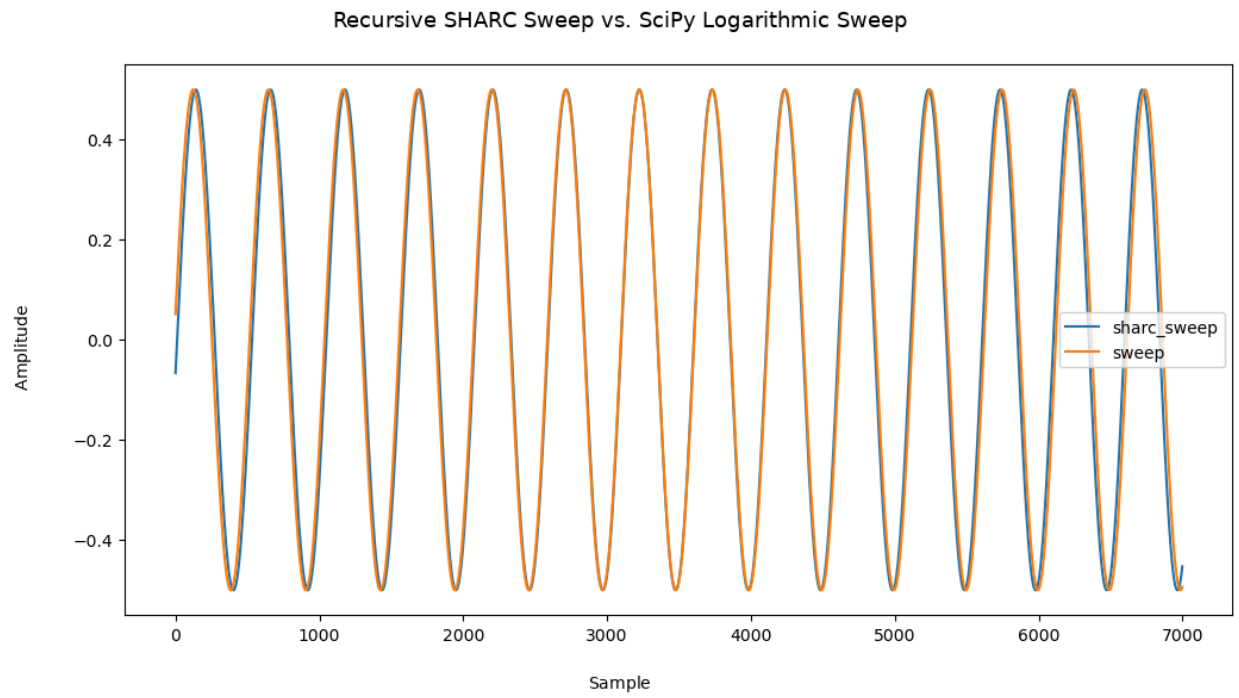


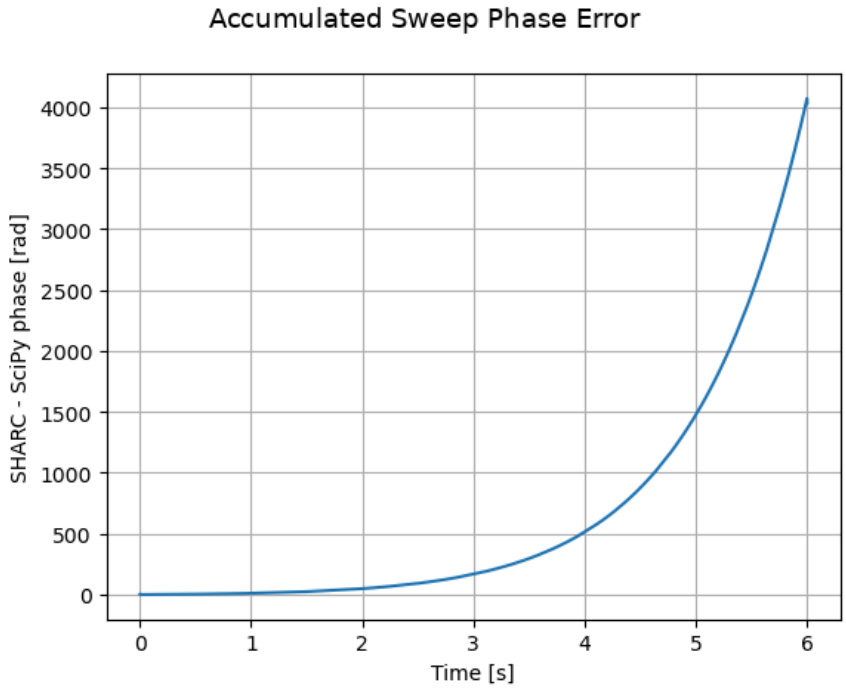
Recorded tones straight into the Scarlett input

- The gain knob on the Scarlett changed the recording amplitude
- With fixed gain knob, the amplitude scaled with the given amplitude

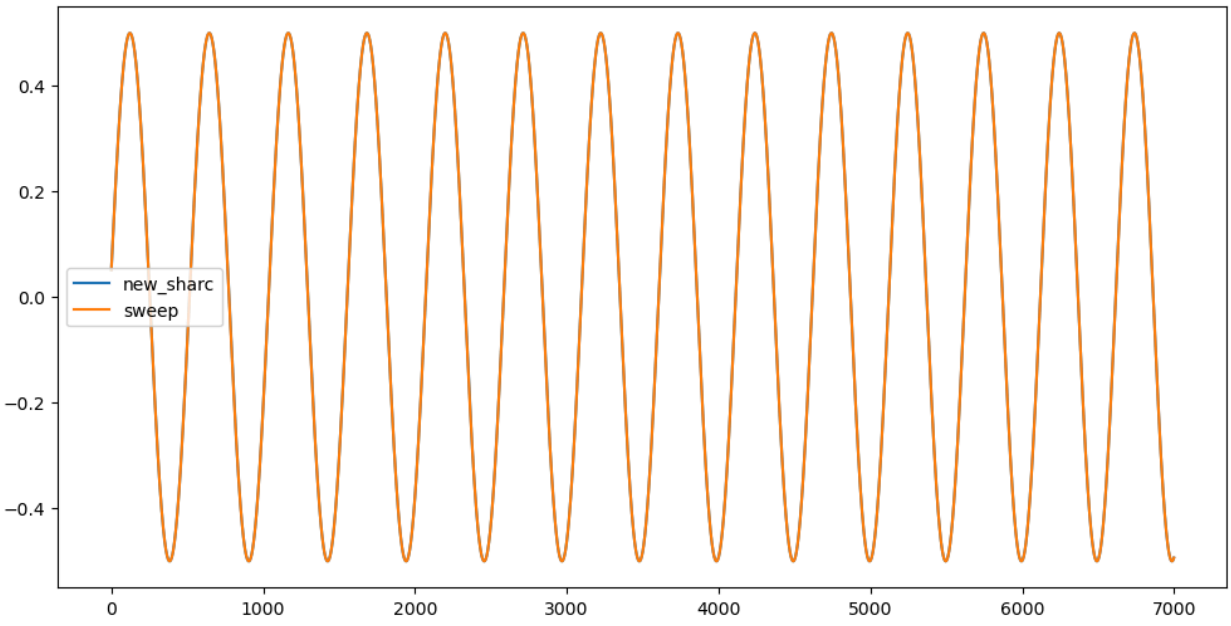
The SHARC sine sweep was wrong

- The phases were different and changed over time

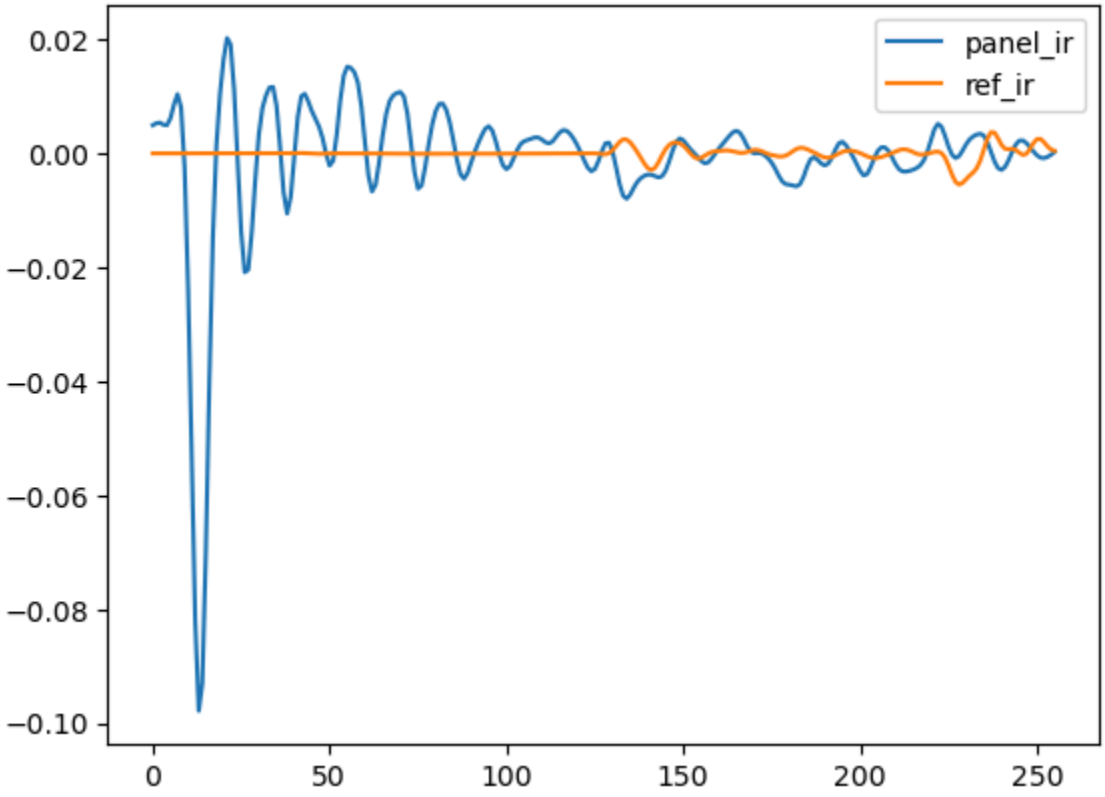


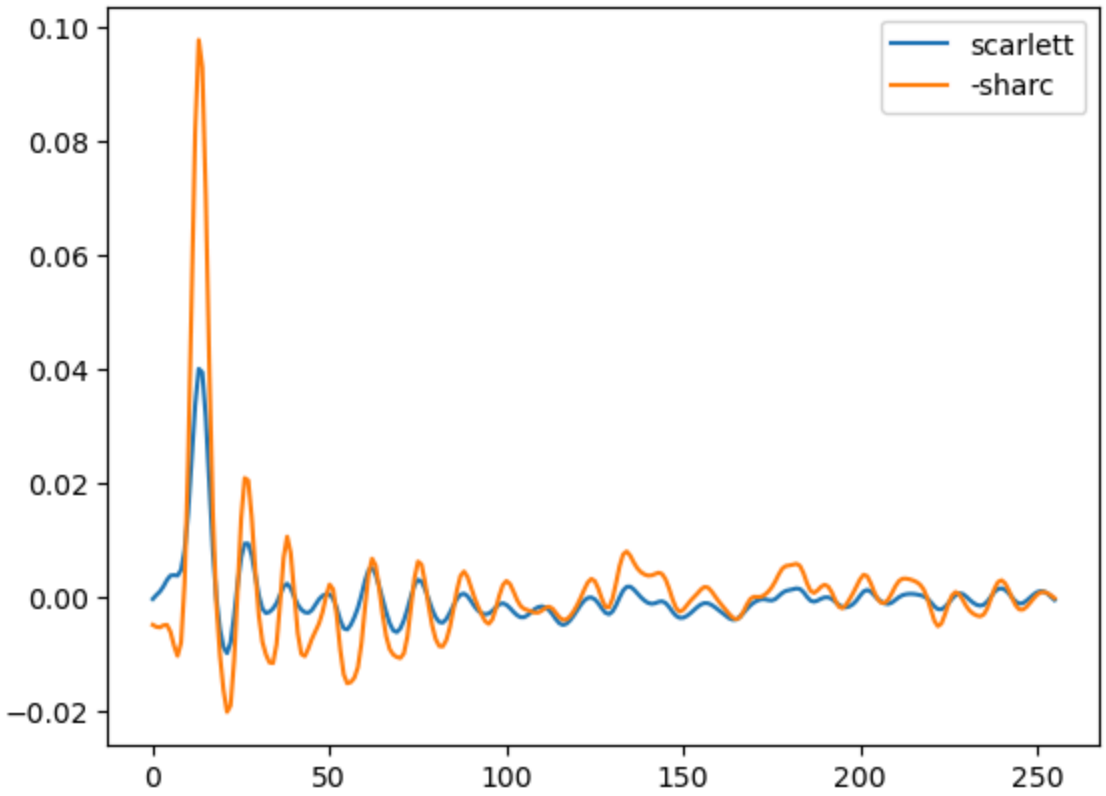


- I was calculating as self multiplication each sample instead of explicitly computing the exponential at each sample
- Small float error accumulated over time
- Computing the closed form for each sample solved the issue



The SHARC IR is still different




scarlett
-sharc
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
0
50
100
150
200
250